

Shaft Winder

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Winder I

- *There are two basic type winder in common.
- *Drum winder, in which rope is stored on drum.
- *Friction winder, in which rope passes over drum (wheel) during hoisting cycle.
- *Drum winder are usually located at some distance from shaft.
- *Drum winder requires a head frame & sheaves to centre hoisting rope in shaft compartment.
- *Friction winder may also be located directly on shaft.
- *Depending up on drum diameter, system may need deflection sheaves to center rope in shaft compartment.

Winder II

- *Ground Mounted Winder
- *Tower Mounted Winder
- *Mine winder are selected & installed at a mine & remains in position for entire life of mine.

Shaft Hoisting System I

EDWARDS

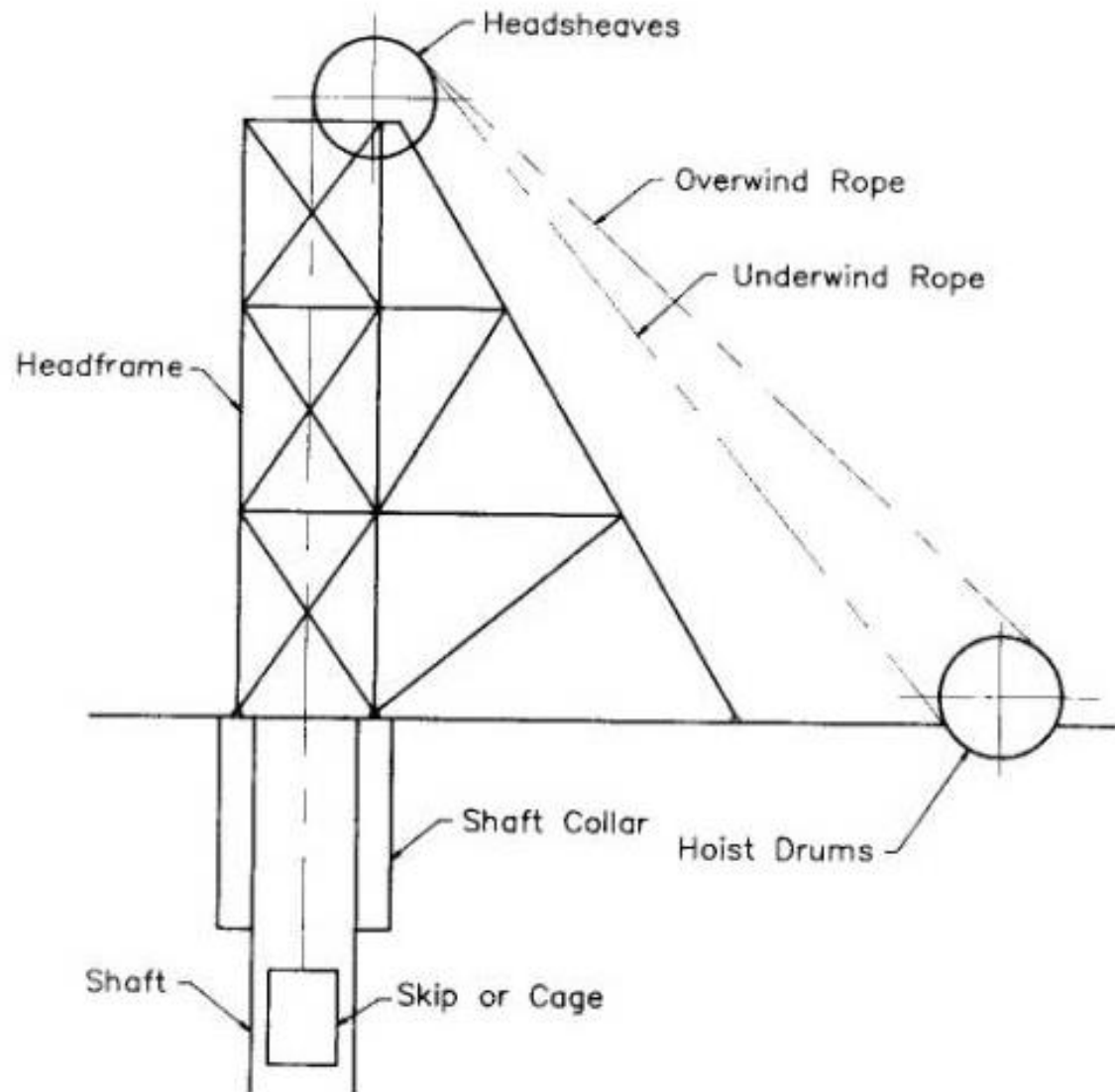
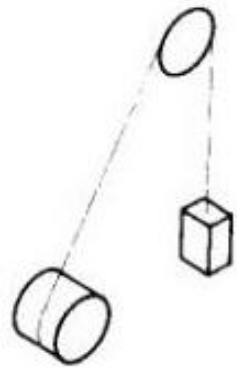


Fig. 17.5.1. Shaft hoisting system.

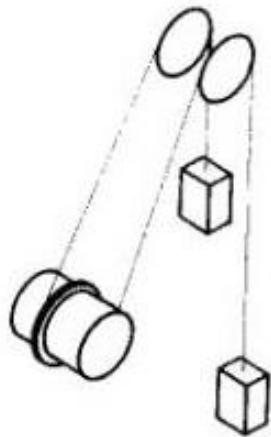
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Various Winder I

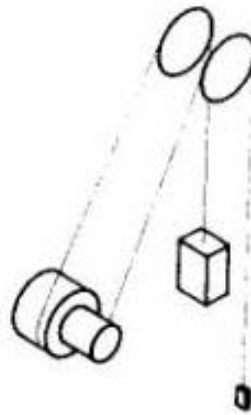
MINING ENGINEER



Single Drum

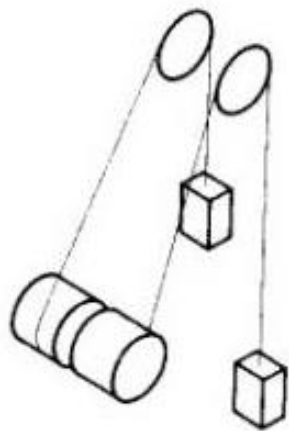


Divided
Single Drum

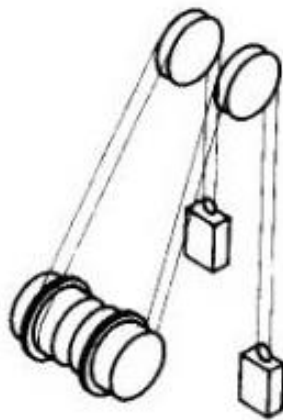


Split
Differential Drum

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Double Drum



Blair



Friction (Koepe)

Fig. 17.5.2. Arrangements of various hoists.

Single Drum Winder I

- *Single drum winder with single rope is a unbalanced winding system.
- *Cost of electric drive becomes quite high for long hoisting distance & high torque.
- *Winder motor must give sufficient torque to handle weight of rope, conveyance & pay load.

Single Drum Double Rope Winder I

- *In balance single drum winding system there are two ropes winded over single drum.
- *One rope is wind on mode & other rope is with wind off mode.
- *When used with skip or cage in balance with counter weight, a single drum winder can serve one or more levels, since location of counter weight is nor important.
- *When used two skip in balance, single drum winder is best used for single level hoisting.
- *Any rope adjustment to locate conveyances is difficult & must be done manually.

Single Drum Double Rope Winder II

- *For shallow shaft, with one layer of rope, no dividing in drum is required.
- *For deeper shaft single drum must be divided to accommodate two ropes for making system balanced.
- *Divided drum can store several layer of rope.
- *Peak horse power is less then with unbalanced hoisting because skip weight & balance weight.
- *Payload & weight of rope is not balanced, maximum unbalanced load occurs, when loaded conveyance is at bottom of shaft.

Split Differential Diameter Drum II

- *It is a single winder available in split differential drum diameter.
- *This type of drum is used with a conveyance & counter weight in balance.
- *If counter weight is wound on smaller diameter drum, it moves less than main conveyance & rope adjustment problem is reduced.

Double Drum, One Drum Clutched I

- *Double drum, one drum clutched is more expansive than single drum winder.
- *This system allows quick adjustment if rope due to initial stretch.
- *As service winder with cage & counter weight, rope changing becomes much easy.
- *As production winder with two skips, rope adjustment to maintain balance hoisting at any level is a malty level operation.

Double Drum, Both Drum Clutched I

- *Double drum winder with both drum clutched has added features of allowing hoisting to continue in one compartment should some thing happens in other compartment.
- *This type of winder are also favored during shaft sinking.

Malty Drum, Blair Winder I

- *In this winder, each conveyance is suspended from two hoist ropes that are each coiled on a drum.
- *This system gives advantage of using smaller diameter rope & drum.
- *This type of winder are used for very deep shaft.

Friction Winder I

- *It consist of with a groove lined with friction material to resist slippage.
- *This winder rope is not attached or stored on drum.
- *This winder can be installed on ground.
- *A single rope was wound around drum & over head sheave to conveyance, in a balance arrangement.
- *In addition, a tail rope of same weight & length as head rope is suspended in shaft below each conveyance.
- *Only out of balance load is pay load.

Friction Winder II

- * As hoisting load gets larger, number of head rope & head sheave increased to a point where it becomes more practice to install hoist on head frame.
- * In order to bring rope centre in line with compartment centre, deflection sheaves must also be installed in head frame below hoist.

Friction & Drum Winder I

- *Double drum winder are perfect winder for shaft sinking.
- *Double drum winder are best choice for hoisting in two compartment from several levels.
- *Drum type winder are better for high pay load from shallow depth.
- *Limitation on dram winder employing a single rope is ultimate strength of rope, because larger rope are difficult to manufacture & handle.
- *Depth capacity of dram winder can be extended by using two ropes per conveyance (Blair Hoist).

Friction & Drum Winder II

- *Blair winder can be used for depth exceeding those of either single rope drum hoist or friction hoist.
- *Friction winder with multiple rope can carry higher pay load & have higher out put per hour then drum winder.
- *Depth range of friction winder can be 1500m (winder travel distance).
- *Friction winder mechanical operation is very simple.
- *Friction winder has low rotational inertia.
- *Friction winder is less costly in compared to drum winder.

Friction & Drum Winder III

- *Friction winder have a lower peak power demand than drum winder with same out put.
- *Friction winder can be operated on relatively light power supply.

Winder Drum I

- *For drum winder, drum must be designed to store required length of rope.
- *Drum should meet statutory requirement concerning fleet angle, rope ratios & must be sufficient strong to withstand bending & crushing force.
- *Drum can be plane faced or grooved.
- *With plain faced drum a section of drum is available for caring live turns of rope.
- *For grooved drum, entire face of drum is available for use.

Winder Drum II

- *Helical grooving is a continuous spiral providing smooth winding with single layer of rope.
- *It can be used with malty layer winding rope.
- *Parallel grooving is made up of individual grooves evenly spaced over drum width.
- *This grooving is used for single layer winding only.
- *Le Bus Grooving is a combination of parallel & helical grooves, thus provide better support.
- *It is most suited for malty layer winding as it reduces rope whip at cross over points.

Winder Drum III

- *For friction winder, drum must be sized to meet statutory requirement for rope to drum ratio.
- *Drum should be wide enough to carry required number of ropes.
- *Tread of friction hoist drum is lined with friction material to resist slippage.
- *Polyurethane, PVC or combination blocks are used as friction insert in to winder drum.

Winder Bearing I

- *Two type of bearing is used in wider.
- *Journal (sliding) bearing.
- *Roller Bearing.
- *Sliding bearing are those in which surface are in sliding contact.
- *Supported members runs on a cylinder, conical or flat.
- *Roller bearing have surface that are in rolling contact & supported member run on hardened steel balls or rollers.

Winder Bearing II

- *In some hoist sliding bearing are used with babbitte lining at drum shaft, pinions & head sheaves.
- *During operation bearing surface are separated by an unbroken film of oil to reduce friction.
- *Now, roller or ball bearing are becoming common
- *Because economy of space, reduction in friction, (low co-efficient of friction) practically eliminating wear due to roller point contact, maintenance of accurate alignment, there consistency in design & manufacture.

Winder Bearing III

*For winder, roller bearing having life of 3,00,000 hours are selected.

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Winder Gearing I

- *Winders may be driven by either ac or dc electric motor.
- *Depending upon hoisting speed, these motors may be connected to shaft directly or through gear drive.
- *Low speed motor may be connected to shaft directly.
- *High speed motor is connected to winder through a gear reducing drive.
- *With high speed motors (300 to 900 rpm) power is transmitted from motor to hoist through coupling connected to single or multiple step, helical gear reduction drive.

Winder Bearing II

- *For designing & selecting suitable gear drive, following information's are utilized.
- *Size type, speed, location & number of motor.
- *Size, type, speed. torque requirement & location of winder.
- *Type of operating cycle.
- *Physical restriction.

Winder Brake I

- *Braking system is required to decelerate, stop & hold winder drum.
- *This system is accomplished with electrical & mechanical braking system
- *Both braking system must operate both under normal & emergency condition.
- *Electric braking can be through regenerative braking, counter torque braking or dynamic braking.
- *In regenerative braking, motor, when connected to a hoist operating on an over holding load, performs as an induction generator, developing braking torque & returning energy to system.

Winder Brake II

- *With counter torque & dynamic braking, there is no energy returned to system, instead, it is consumed in secondary resistance as heat.
- *During normal operating conditions, mechanical brake serves two purposes, firstly it helps to slow down drum after speed has been greatly reduced electrically & holds drum at rest.
- *Under this conditions, mechanical brake does little work & its design & operating characteristics can be fairly simple & straight forward.

Winder Brake III

- *Under emergency conditions, however, winder must be stopped with mechanical brake as quickly as possible, without damaging hoist, brake & conveyance.
- *There are two types of brake in operation, disc brake & caliper brake (parallel motion brake).
- *Disc brake are mostly used on friction winder; however, they can also be used with drum winder.
- *With caliper brake, two shoes are pressed against periphery of brake drum through pull of a single or double draw bar acting through series of linkages.

Winder Brake IV

- *Sources of force can be a brake weight or nest of spring.
- *When not in use, weight is hold up or spring are hold pressed by hydraulic oil cylinder.
- *When brake is applied, oil flow from cylinder to allow force of weight or spring to apply brake.
- *Control of flow is extremely important as it determine speed & amount of force to be applied.
- *In one type of disc brake, brake is operated via lever system & during emergency stop via a falling weight.

Winder Brake V

- *This system is similar to conventional post brake system.
- *In other disc brake, hydraulically operated spring applied unit are used.
- *This brake units are mounted around brake disc fixed with winder drum.
- *Required number of unit is determined by total braking force required for particular hoist.
- *Advantage of disc brake are there relatively small dimension, light weight & easy replacement.

Winder Clutch I

- *Drum winder used for unbalanced hoisting or when operating from several levels, should have at least one of drum clutch connected to drum shaft.
- *With such a hoist, clutched drum is supported in shaft through sleeve bushing.
- *Clutch is used to transport motor torque to drum.
- *In order to prevent drum from rotating when it is unclutched, it is necessary that clutch operating mechanism & drum brake be interlocked.
- *Clutch is operated through a two, three or four spider that is attached to turning drum shaft.

Winder Clutch II

- *This arms are perpendicular to shaft & shifted parallel to axis of drum shaft.
- *A clutch ring with matching teeth on its entire periphery is bolted to inside of drum.
- *As spider is moved, these two sets of teeth are free to engage at any point on circumference of drum.
- *Driving torque is then transmitted from clutch to drum through bolts used to hold clutch ring to drum flange.

Winder Motor I

- *Winder can be powered by Electric, steam, compressed air, internal combustion engine.
- *Electric power is most common in winder system.
- *Choice of electric motor drive lies between alternating current motor & direct current motor.
- *Direct current being supplied from either motor generator set or silicon control rectifier.
- *Principle factor to be considered in selecting a motor are matching configuration, electric power system & economics.

Winder Motor II

- *AC motor has advantage if heaving lowest initial cost & can use power normally supplied.
- *AC motor has higher starting torque & difficult to automate.
- *AC motor generally require gear train to drive hoist.
- *In present technology VVVF AC motor is able to over come all these problem & is getting more acceptance as winder drive.
- *AC motor requires very low maintenance in compared to DC system.

Winder Motor III

- *DC motor provides accurate & sensitive control, is easily automated & has a low starting torque.
- *DC motor may be directly connected to hoist, these eliminates gear drive & space requirement of such drive.
- *Choice of power conversion equipment is effected by quality of power system available, ratio of hoisting load (electrical) & total plant load, personal preference.

Winder Control I

- *Winder control systems are required to monitor speed & location of each conveyance moving in shaft.
- *When ever a conveyance exceeds a resettled safe speed distance profile, control system initiate an action to prevent moving conveyance from striking a permanent obstruction, regardless of its speed & direction of travel.
- *In addition, control system may be used to shut down equipment due to high operating temperature, excess brake wear, rope slip, slack rope, loss of power etc
- *Control system for friction winder are much more numerous & complex than for drum hoist.

Winder Control II

*Most important factor to examine while selecting control system are simplicity, reliable & ease of adjustment, ease of maintenance.

Winder Selection I

- *Size of winder is expressed by drum dimensions, horse power rating of motor & rope pull.
- *Drum winder & friction winder consists of two separate machines, each of which must be designed to produce desired result.
- *Winder consist of following mechanical parts, drum, drum shaft, gear, brakes,
- *Electrical parts consists of drive motor & controls.
- *Mechanical parts of winder is designed to support hoisting rope & its load.

Winder Selection II

- *Electrical portion of hoist is designed to provide sufficient torque to turn drum & raise or lower rope, conveyance & payload.
- *To design & build a hoist following basic information are required.
- *Hoisting distance, production rate (t per hour), maximum load, type of guide (wood, steel, rope).
- *For a given depth & production rate, there is an optimum load that results in lowest cost for hoist.
- *It is important for selected hoist to have some excess capability for drum & motor.

Winder Selection III

- *This shall allow some changes in future.
- *Following information must be considered before selecting, designing mine winder.
- *Ultimate use of shaft, men, material, ventilation, exploration, proposed phasing of expansion. Surface plant lay out, head frame, bin collection.
- *Underground layout (level, station, loading pocket)
- *Type of power distributions system.
- *To design a winding system, thesis basic information are to be considered.

Winder Selection IV

- *Production (hoisting) rate t per hour.
- *Maximum load to be hoisted.
- *Hoist distance,
- *Weight of pay load & conveyance.
- *Diameter of hoisting rate.

Winder Duty Cycle I

- *Duty cycle describes total time it takes to move a conveyance from bottom of its wind to top.
- *Duty cycle often depicts graphically as a time speed diagram.
- *Duty cycle must include periods of time for initial creep, acceleration, full speed, retardation, dumping, loading rest.
- *Physical law of uniformly accelerated motion are used to determine time required & distance travelled once value of acceleration rate, final velocity & shaft depth are known.

Winder Duty Cycle II

- *Maximum velocity of skip / conveyance should not exceed following value for different type of guides.
- *Wooden Guide 10m/second
- *Steel guide 15m/second
- *Rope guide 20m/second.
- *There is a limit for acceleration & retardation.
- *Limit is normally 0.7m/sec sq.

Winder Production Rate I

- *Required production rate (hoisting rate) for skip is expressed in terms of average ton per hour hoisted.
- *If mill operates on different schedule to that of mine, production rate must be adjusted to eliminate difference between these rates.
- *It is better to have extra hoisting capacity in shaft, so that mill does not starve for ore from mine.
- *In addition, moisture content of ore must be taken in to consideration & included in quantity of material to be hoisted.

Winder Production Rate II

- *At this stage, concept of hoist utilization factor should be introduced.
- *In certain well-organized shaft, utilization factor can be as high as 0.92, but more normal figure is 0.7

Winder Skip Size I

*After cycle time & production rate are known, pay load requirements for skip are derived by

$$P = Pr / T = Pr / C * 3600$$

P > pay load of skip

Pr > production rate ton/hour.

T > number of trip per hour

C > cycle time in second.

3600 is 3600 second / hour.

Winder Skip Size II

*Approximate weight of skip can be determined from

$$W = 0.5 + 680 \text{ in kg}$$

$W > \text{pay load of skip}$

*Roughly, skip weight is $\frac{3}{8}$ pay load.

Winder Rope Size I

- *Correct rope size is important for winding system.
- *Selection of rope must fulfill statutory requirement, like safety factor, rope & drum diameter ratio, length of rope etc.
- *Type of rope is selected by duty of rope in winding system.
- *Diameter of rope is selected based on weight of rope, pay load, skip load, maximum rope length suspended in shaft, number of rope.

Winder Drum Size I

- *Having determined rope diameter, drum diameter can be decided.
- *As per Indian Regulation, ratio of drum diameter & rope diameter is 1:100.
- *Drum width shall fulfill requirement of fleet angle.
- *Drum size to be able to hold extra rope as per regulation, plus rope of sufficient to perform destructive test every 6 months.
- *Winder drum should have sufficient flange height to prevent rope jump out.

Number of Winder Required I

- *Normally shaft is designed to be multipurpose.
- *In one shaft, there can be number of winders for ore, waste, personnel, material movement, maintenance etc
- *A shaft is equipped with more than one winder to serve this purposes.
- *It is required to list all necessarily & its duration.
- *Following information's are required to decide number of winders to be fitted in shaft.
- *Number of hoisting trip for ore, waste, material, men.

Number of Winder Required II

- *Time required for hoist, shaft, rope & conveyance inspection as required by law & good maintenance practices.
- *Time required for internal movement of personnel & material.

Ore Hoisting I

- *In order to determine number of hours per day required to hoist ore following information's are required.
- *Planned tonnage to be hoisted daily.
- *Production rate for skip hoist in ton per hour.
- *Normally a winder is designed to hoist ore to be hoisted in day in 12 hours.
- *Balance time is available for maintenance & other hindrances.
- *Hoisting capacity of winder is designed accordingly.

Waste Hoisting I

- *Waste hoisting time is calculated in way of ore hoisting.
- *In waste hoisting, winder utilization time is taken about 70%.
- *When calculating daily waste production, consideration should be given to use waste in underground as back fill.
- *This shall reduce daily waste hoisting requirement.

Material & Supply I

- *In order to determine number of hours per day requirement to lower / hoist materials & supplies.
- *Following information's are to be known for determining winder load.
- *Number of materials & suppliers trip per day.
- *Hoisting duty cycle for material lowering & hoisting.

Personnel Hoisting I

- *Projected daily employee requirement that must be lowered & raised through hoisting facilities is either known or calculated using daily manpower data, hoisting time required for transporting personal to & from under ground is calculated after determining cycle time..
- *In estimated cycle time, consideration is given to followings.
- *Cage capacity.
- *Number of working levels underground to which personnel are to be transported.

Personnel Hoisting II

- *Hoisting speed to first level & then in between two levels.
- *Loading & unloading time (one level).
- *Idle time (per cycle).

Maintenance & Inspection I

- *Statutory maintenance, inspection & test required as regulated by authority to be carried out.
- *In order to allow maximum time for ore hoisting, it is normal practice at most major mine to carry out maintenance & repair of a major nature on weak end when requirement for lowering personnel & material is at minimum.
- *Some time repair or maintenance may have to be carried out in working days for safe operation of winding system, shaft & mine.

Internal Hoisting I

- *In some mining operation it is necessary to transfer personnel, material & equipments between levels using shaft hoisting equipment.
- *Work required to carry out these task is termed as internal hoisting.
- *Time required to complete these type of hoisting can best be estimated through personal experience or by discussion with operator of similar mine.

Idle Time I

- *It is not possible to plan to operate winding system 100% of time.
- *Some idle time must be provided in any hoisting schedule.
- *Amount of time to be alloyed is generally a matter of experience on part of designer.
- *Total operating time scheduled during planning stage should not exceed 70% of total operating time available, that is 17 roughly hours in a day.

Reference I

*SME Mining Engineering Hand Book, 2nd Edition,
Volume 2 Page 1646 to 1652